

Supervised and Unsupervised Machine Learning

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Unit Overview: Supervised Learning Algorithms

In this unit, we will study the following **supervised learning algorithms**:

- **Linear Regression**: Predicts continuous values using best-fit line.
- **Logistic Regression**: Classification using probabilistic decision boundary.
- **k-Nearest Neighbours (k-NN)**: Classifies based on nearest data points.
- **Decision Trees**: Rule-based hierarchical splitting of data.
- **Naïve Bayes Classifier**: Probabilistic model using Bayes theorem.
- **Support Vector Machine (SVM)**: Maximum margin classifier for optimal separation.

Unit Overview: Unsupervised Learning Algorithms

We will also study the following **unsupervised learning algorithms**:

- **K-means Clustering**: Groups data into K clusters based on similarity.
- **Hierarchical Clustering**: Builds tree-like structure of clusters.
- **DBSCAN**: Density-based clustering for arbitrary shaped clusters.
- **Anomaly Detection**: Identifies rare or unusual patterns in data.

What is Linear Regression?

- Linear Regression is a supervised learning algorithm.
- It predicts continuous numerical values.
- It models the relationship between input and output variables using a straight line.

Examples

- House price prediction
- Salary estimation
- Sales forecasting
- Temperature prediction

Simple Linear Regression

Simple Linear Regression uses one independent variable to predict one dependent variable.

$$y = \beta_0 + \beta_1 x$$

where

- x : Independent variable
- y : Dependent variable
- β_0 : Intercept
- β_1 : Slope (coefficient)

Example Dataset

Area (sq ft)	Price (\$1000)
1000	200
1500	280
2000	360
2500	430

As the area increases, the house price also increases.

Components of Regression Equation

$$y = 50 + 3x$$

Intercept (β_0)

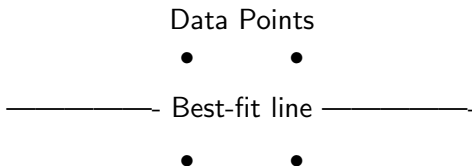
- Value of y when $x = 0$.

Slope (β_1)

- Change in y for one unit increase in x .

Goal of Linear Regression

The objective is to find the best fitting line that minimizes prediction error.



Prediction Error

Predicted value:

$$\hat{y}$$

Residual (Error):

$$Error = y - \hat{y}$$

Residuals measure the difference between actual and predicted values.

Cost Function

Linear Regression minimizes Mean Squared Error (MSE).

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Smaller MSE indicates better performance.

Multiple Linear Regression

Multiple Linear Regression uses more than one feature.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

Examples of features:

- Area
- Number of bedrooms
- Age of house
- Location

Assumptions of Linear Regression

- 1 Linear relationship between variables.
- 2 Independent observations.
- 3 Constant variance of errors.
- 4 Normally distributed residuals.
- 5 No multicollinearity.
- 6 No significant outliers.

Evaluation Metrics

Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{MSE}$$

Coefficient of Determination

$$R^2$$

Coefficient of Determination (R^2)

R^2 measures how well the regression model explains the variability in the data.

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

where

- $SS_{res} = \sum (y_i - \hat{y}_i)^2$: Residual Sum of Squares
- $SS_{tot} = \sum (y_i - \bar{y})^2$: Total Sum of Squares
- $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$: Mean of the actual values

Interpretation

- $R^2 = 1$: Perfect fit
- $R^2 = 0$: Model explains none of the variance
- Higher R^2 indicates a better model fit

Advantages

- Simple and easy to understand.
- Computationally efficient.
- Easy to interpret.
- Works well for linear relationships.

Limitations

- Sensitive to outliers.
- Assumes linearity.
- Performs poorly on nonlinear data.
- Can be affected by multicollinearity.

Applications

- House price prediction
- Sales forecasting
- Healthcare analytics
- Economic modeling
- Demand estimation
- Risk assessment

Summary

- Linear Regression predicts continuous values.
- It models the relationship between variables using a straight line.
- The objective is to minimize prediction error.
- Common evaluation metrics include MAE, MSE, RMSE, and R^2 .
- It is one of the most widely used machine learning algorithms.

What is Logistic Regression?

- Logistic Regression is a supervised learning algorithm.
- Used for classification problems.
- Predicts the probability that an instance belongs to a class.
- Most commonly used for binary classification.

Examples

- Spam detection
- Disease diagnosis
- Credit card fraud detection
- Customer churn prediction

Binary Classification

Two possible outcomes:

Positive Class (1)

Negative Class (0)

Examples:

- Yes / No
- True / False
- Spam / Not Spam
- Pass / Fail

Why Not Linear Regression?

Linear regression may predict values outside the interval $[0, 1]$.

Logistic regression converts predictions into probabilities between 0 and 1.

$$0 \leq P(Y = 1|X) \leq 1$$

Hence, it is suitable for classification tasks.

The Sigmoid Function

Logistic regression uses the sigmoid function:

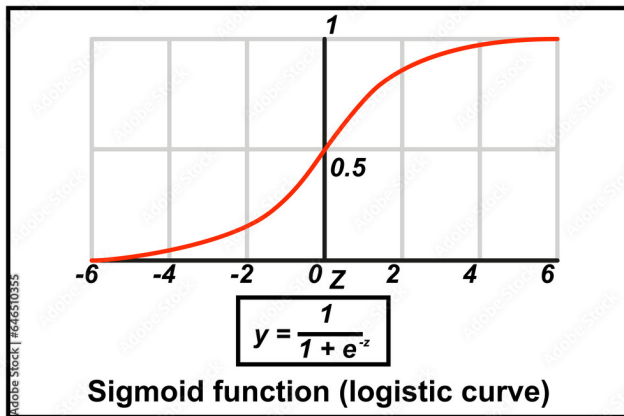
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

where

$$z = \beta_0 + \beta_1 x$$

The output is always between 0 and 1.

Shape of the Sigmoid Function



- S-shaped curve.
- Converts any real number into a probability.

Decision Boundary

Predicted probability:

$$P(Y = 1|X)$$

Classification rule:

$$\hat{y} = \begin{cases} 1, & P(Y = 1|X) \geq 0.5 \\ 0, & P(Y = 1|X) < 0.5 \end{cases}$$

Threshold value is usually 0.5.

Log Odds (Logit)

The logit function is

$$\log \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 x$$

where

- p is the probability of belonging to class 1.
- $\frac{p}{1-p}$ represents the odds.

Cost Function

Cross-entropy is a loss function used to measure how different the predicted probabilities are from the true labels. Logistic regression uses Binary Cross-Entropy Loss:

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Goal:

Minimize the loss function.

Training Process

- 1 Collect training data.
- 2 Initialize parameters.
- 3 Compute predictions.
- 4 Calculate loss.
- 5 Update parameters using gradient descent.
- 6 Repeat until convergence.

Common evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC Curve
- AUC

Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

- TP: True Positive
- TN: True Negative
- FP: False Positive
- FN: False Negative

Performance Metrics

Common evaluation metrics:

- **Accuracy**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision**

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall (Sensitivity)**

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1-Score**

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **ROC Curve**

ROC plots: TPR vs FPR

where

$$TPR = \frac{TP}{TP + FN}, \quad FPR = \frac{FP}{FP + TN}$$

Advantages

- Simple and efficient.
- Easy to interpret.
- Produces probability estimates.
- Works well for linearly separable data.

Limitations

- Assumes linear relationship between features and log odds.
- Sensitive to outliers.
- May perform poorly on complex nonlinear data.
- Requires sufficient training data.

Applications

- Email spam detection
- Medical diagnosis
- Credit scoring
- Fraud detection
- Sentiment analysis
- Customer churn prediction

Summary

- Logistic regression is used for classification.
- It predicts probabilities using the sigmoid function.
- Output values lie between 0 and 1.
- Binary cross-entropy is used as the loss function.
- Performance is evaluated using accuracy, precision, recall, F1-score, ROC, and AUC.

Support Vector Machine (SVM): Conceptual Foundation

Support Vector Machine (SVM) is a powerful **supervised learning algorithm** used for both classification and regression, though it is most commonly applied to classification problems.

Core Idea:

- SVM finds an **optimal separating hyperplane** that divides data points of different classes.
- The optimality is defined by the **maximum margin principle**.
- The margin is the distance between the hyperplane and the nearest data points from each class.

Key Principle: Maximum Margin

- Among all possible separating hyperplanes, SVM selects the one that maximizes the margin.
- This improves **generalization ability** and reduces overfitting.

Support Vectors:

- The closest data points to the decision boundary.
- These points are critical because they fully determine the position of

Support Vector Machine (SVM): Geometric View

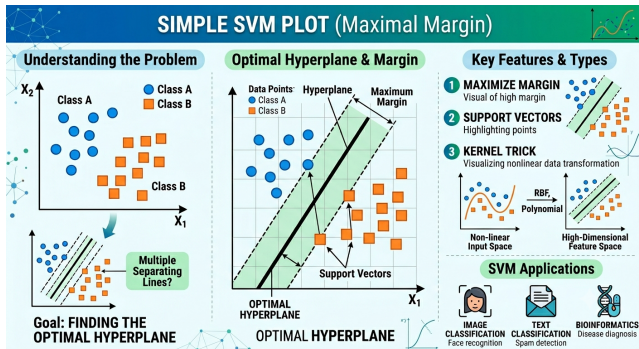


Figure: SVM: maximum margin hyperplane, margins, and support vectors

Key Idea:

- The hyperplane separates two classes.
- Margins define the maximum separation region.
- Support vectors are the closest points that define the boundary.

Support Vector Machine (SVM): Intuition

SVM is a **maximum margin classifier** that finds the best separating hyperplane between two classes.

- Goal: Find hyperplane $w^T x + b = 0$
- Maximize the margin between two classes
- Support vectors are the closest points that define the boundary

Decision boundary:

$$w^T x + b = 0$$

Margins:

$$w^T x + b = +1 \quad \text{and} \quad w^T x + b = -1$$

SVM Optimization Problem

Support Vector Machine is formulated as a **convex optimization problem** that finds the optimal separating hyperplane with maximum margin.

Hard Margin SVM (Linearly Separable Data):

$$\begin{aligned} \min_{w,b} \quad & \frac{1}{2} \|w\|^2 \\ \text{subject to} \quad & y_i(w^T x_i + b) \geq 1, \quad \forall i \end{aligned}$$

Key Idea:

- Minimizing $\|w\|^2$ maximizes the margin
- Constraints enforce correct classification

Soft Margin SVM (Practical Case)

For non-linearly separable or noisy data, SVM introduces slack variables ξ_i .

$$\begin{aligned} \min_{w, b, \xi} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \\ \text{subject to} \quad & y_i(w^T x_i + b) \geq 1 - \xi_i, \\ & \xi_i \geq 0, \quad \forall i \end{aligned}$$

Key Idea:

- ξ_i allows margin violation / misclassification
- C controls trade-off between margin and error
- Large $C \rightarrow$ low bias, high variance
- Small $C \rightarrow$ wider margin, more tolerance

SVM: Margin Concept

The geometric margin is:

$$\text{Margin} = \frac{2}{\|w\|}$$

Key Insight:

- Maximizing margin $\Leftrightarrow$ minimizing $\|w\|$
- Larger margin improves generalization

Optimization Insight (Why SVM Works)

Why this works:

- Objective is **convex** $\rightarrow$ global optimum guaranteed
- Solution depends only on **support vectors**
- Solved using:
 - Lagrange multipliers
 - Quadratic Programming (QP)
 - SMO (Sequential Minimal Optimization)

Key Insight:

Maximize margin $\Leftrightarrow$ Minimize $\|w\|$

Kernel Trick: Motivation

Problem:

- Some datasets are not linearly separable in original space

Idea:

$\phi(x) : x \rightarrow$ higher-dimensional feature space

Goal:

- Make data linearly separable in transformed space

Challenge: Explicit computation of $\phi(x)$ is expensive.

Kernel Trick: Core Idea

Instead of computing $\phi(x)$ explicitly:

$$K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$$

Key Idea:

- Compute inner product in high-dimensional space implicitly
- Avoid explicit transformation

Result:

- Nonlinear boundary in original space
- Linear separation in feature space

Kernel Trick: Intuition Example

Non-linearly separable data:

- Class +1: points inside circle
- Class -1: points outside circle

Feature mapping:

$$\phi(x_1, x_2) = (x_1^2, x_2^2, x_1 x_2)$$

Idea: Data becomes linearly separable in higher dimension.

Common Kernel Functions

Linear Kernel

$$K(x_i, x_j) = x_i^T x_j$$

Polynomial Kernel

$$K(x_i, x_j) = (x_i^T x_j + c)^d$$

RBF (Gaussian) Kernel

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

Interpretation:

- Linear $\rightarrow$ simple boundary
- Polynomial $\rightarrow$ curved boundary
- RBF $\rightarrow$ highly flexible nonlinear boundary

$$f(x) = \sum_{i \in SV} \alpha_i y_i K(x_i, x) + b$$

Key Points:

- Only support vectors contribute
- Kernel replaces dot product
- Enables nonlinear classification efficiently

Main Idea: Kernel trick enables high-dimensional learning without explicit computation.

SVM Training Procedure (Intuition)

Initialize:

$$w = 0, \quad b = 0$$

For each (x_i, y_i) :

$$y_i(w^T x_i + b)$$

Case 1: Correct classification (≥ 1)

$$w \leftarrow w - \eta(2\lambda w)$$

Case 2: Misclassified (< 1)

$$w \leftarrow w - \eta(2\lambda w - Cy_i x_i)$$

$$b \leftarrow b + \eta y_i$$

SVM Classification Example

Consider two classes:

$$+1 : (3, 3), (4, 3)$$

$$-1 : (1, 1), (2, 1)$$

Assume the separating hyperplane is

$$w = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad b = -4$$

Decision function:

$$f(x) = w^T x + b$$

Class prediction:

$$\hat{y} = \text{sign}(f(x))$$

SVM Classification Example

For point (3, 3):

$$f(3, 3) = 1(3) + 1(3) - 4 = 2$$

$$\hat{y} = +1$$

For point (1, 1):

$$f(1, 1) = 1(1) + 1(1) - 4 = -2$$

$$\hat{y} = -1$$

Since the signs match the true labels, the points are correctly classified.

Distance from Hyperplane

Distance of a point from the hyperplane:

$$d = \frac{|w^T x + b|}{\|w\|}$$

For point (3, 3):

$$d = \frac{|2|}{\sqrt{1^2 + 1^2}} = \frac{2}{\sqrt{2}} = 1.414$$

For point (1, 1):

$$d = \frac{|-2|}{\sqrt{2}} = 1.414$$

Both points are equally distant from the boundary.

Margin Calculation

Suppose the support vectors satisfy

$$w^T x + b = 1$$

and

$$w^T x + b = -1$$

The margin is

$$\text{Margin} = \frac{2}{\|w\|}$$

Since

$$\|w\| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

we obtain

$$\text{Margin} = \frac{2}{\sqrt{2}} = \sqrt{2} = 1.414$$

What Does SVM Optimize?

SVM searches for w and b that maximize the margin.

$$\max \frac{2}{\|w\|}$$

Equivalent optimization:

$$\min \frac{1}{2} \|w\|^2$$

subject to

$$y_i(w^T x_i + b) \geq 1$$

Key Idea:

A larger margin generally leads to better generalization on unseen data.

k-Nearest Neighbors (k-NN)

- k-NN is a supervised learning algorithm.
- Used for classification and regression.
- Classifies a new instance based on its nearest neighbors.
- It is a non-parametric and instance-based learning method.

How k-NN Works

- 1 Choose the value of k .
- 2 Compute the distance between the test point and training points.
- 3 Select the k nearest neighbors.
- 4 Assign the majority class.

"Similar instances belong to similar classes."

Distance Measures

Euclidean distance:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Manhattan distance:

$$d(x, y) = \sum_{i=1}^n |x_i - y_i|$$

Commonly used distance metrics:

- Euclidean Distance
- Manhattan Distance
- Minkowski Distance

Choosing the Value of k

- Small k :
 - Sensitive to noise
 - May lead to overfitting
- Large k :
 - Smoother decision boundary
 - May lead to underfitting

Common values:

$$k = 3, 5, 7$$

Advantages and Limitations of k-NN

Advantages

- Simple and easy to implement.
- No training phase.
- Works well for small datasets.

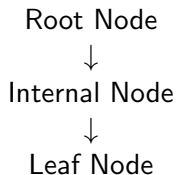
Limitations

- Computationally expensive.
- Sensitive to irrelevant features.
- Requires feature scaling.

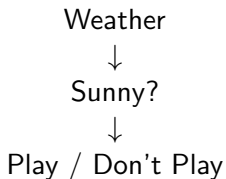
Decision Trees

- A supervised learning algorithm.
- Used for classification and regression.
- Represents decisions as a tree structure.
- Internal nodes represent tests on features.
- Leaf nodes represent class labels.

Structure of a Decision Tree



Example:



Entropy

Entropy measures impurity in a dataset.

$$Entropy(S) = - \sum_{i=1}^c p_i \log_2(p_i)$$

where

- p_i = probability of class i
- c = number of classes

Lower entropy indicates higher purity.

Information Gain measures the reduction in entropy.

$$IG(S, A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

The attribute with the highest information gain is selected for splitting.

Advantages and Limitations of Decision Trees

Advantages

- Easy to interpret.
- Requires little data preprocessing.
- Handles numerical and categorical data.

Limitations

- Can overfit training data.
- Sensitive to small changes in data.
- Trees may become complex.

Naïve Bayes Classifier

- Probabilistic supervised learning algorithm.
- Based on Bayes' theorem.
- Assumes features are conditionally independent.
- Efficient for high-dimensional data.

Bayes' Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

where

- $P(A|B)$: Posterior probability
- $P(B|A)$: Likelihood
- $P(A)$: Prior probability
- $P(B)$: Evidence

Naïve Bayes Classification Rule

For features $X = (x_1, x_2, \dots, x_n)$:

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$$

Prediction:

$$\hat{C} = \arg \max_C P(C|X)$$

Choose the class having the highest posterior probability.

Types of Naïve Bayes

- Gaussian Naïve Bayes
- Multinomial Naïve Bayes
- Bernoulli Naïve Bayes

Applications:

- Spam filtering
- Sentiment analysis
- Text classification
- Document categorization

Advantages and Limitations of Naïve Bayes

Advantages

- Simple and fast.
- Performs well with large datasets.
- Effective for text classification.

Limitations

- Assumes feature independence.
- Zero-frequency problem may occur.
- Performance decreases when features are correlated.

Summary

Property	k-NN	Decision Tree	Naïve Bayes
Type	Instance-based	Tree-based	Probabilistic
Training Speed	Fast	Moderate	Very Fast
Interpretability	Low	High	Moderate
Handles Nonlinearity	Yes	Yes	Limited
Suitable for Text Data	No	Moderate	Excellent

Unsupervised Learning

- Unsupervised learning deals with unlabeled data.
- The goal is to discover hidden patterns and structures in the data.
- No target variable is provided.

Common tasks:

- Clustering
- Dimensionality reduction
- Association rule mining
- Anomaly detection

Clustering

Clustering groups similar data points together.

Applications:

- Customer segmentation
- Image segmentation
- Document clustering
- Gene expression analysis

K-Means Clustering

- Partition-based clustering algorithm.
- Divides data into K clusters.
- Each cluster is represented by its centroid.
- Data points are assigned to the nearest centroid.

K-Means Algorithm

- 1 Select the number of clusters K .
- 2 Initialize K centroids.
- 3 Assign each point to the nearest centroid.
- 4 Recompute cluster centroids.
- 5 Repeat until convergence.

Objective Function of K-Means

K-Means minimizes the within-cluster sum of squares:

$$J = \sum_{i=1}^K \sum_{x_j \in C_i} \|x_j - \mu_i\|^2$$

where

- C_i : Cluster i
- μ_i : Centroid of cluster i

Advantages and Limitations of K-Means

Advantages

- Simple and efficient.
- Fast on large datasets.
- Easy to implement.

Limitations

- Number of clusters must be specified.
- Sensitive to outliers.
- Assumes spherical clusters.

Hierarchical Clustering

- Creates a hierarchy of clusters.
- Does not require specifying the number of clusters initially.
- Produces a tree-like structure called a dendrogram.

Types:

- Agglomerative (Bottom-Up)
- Divisive (Top-Down)

Agglomerative Clustering

Algorithm:

- 1 Start with each point as a separate cluster.
- 2 Merge the closest clusters.
- 3 Repeat until only one cluster remains.

Distance measures:

- Single linkage
- Complete linkage
- Average linkage

Advantages and Limitations of Hierarchical Clustering

Advantages

- No need to predefine the number of clusters.
- Provides dendrogram visualization.

Limitations

- Computationally expensive.
- Sensitive to noise and outliers.

Density-Based Spatial Clustering of Applications with Noise (DBSCAN).

- Density-based clustering algorithm.
- Groups closely packed points.
- Detects noise and outliers.
- Can discover clusters of arbitrary shapes.

Key Parameters of DBSCAN

ϵ (Epsilon)

- Radius around a point.

MinPts

- Minimum number of neighboring points required to form a cluster.

Point categories:

- Core points
- Border points
- Noise points

Advantages and Limitations of DBSCAN

Advantages

- Handles arbitrary-shaped clusters.
- Robust to outliers.
- No need to specify the number of clusters.

Limitations

- Sensitive to parameter selection.
- Struggles with varying densities.

Anomaly Detection

Anomaly detection identifies unusual observations that differ significantly from normal data.

Examples:

- Credit card fraud detection
- Network intrusion detection
- Medical diagnosis
- Manufacturing fault detection

Types of Anomalies

- 1 Point Anomalies
- 2 Contextual Anomalies
- 3 Collective Anomalies

Example:

- Unusual credit card transaction.
- Sudden increase in network traffic.

Common Approaches for Anomaly Detection

- Statistical methods
- Distance-based methods
- Density-based methods
- Clustering-based methods
- Machine learning methods

Examples:

- Isolation Forest
- One-Class SVM
- Local Outlier Factor (LOF)

Comparison of Clustering Algorithms

Property	K-Means	Hierarchical	DBSCAN
Need K beforehand	Yes	No	No
Handles Outliers	Poor	Moderate	Good
Arbitrary Shapes	No	Limited	Yes
Scalability	High	Low	Moderate
Noise Detection	No	No	Yes

Summary

- Unsupervised learning works with unlabeled data.
- K-Means partitions data into K clusters.
- Hierarchical clustering creates a dendrogram of nested clusters.
- DBSCAN discovers density-based clusters and detects noise.
- Anomaly detection identifies unusual patterns and outliers.